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Bounded capabilities: what exactly is an agent authorized to call?

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For your favorite Tax API, which one do you prefer ?
« Q: Which form do I need to report this income type? »

A

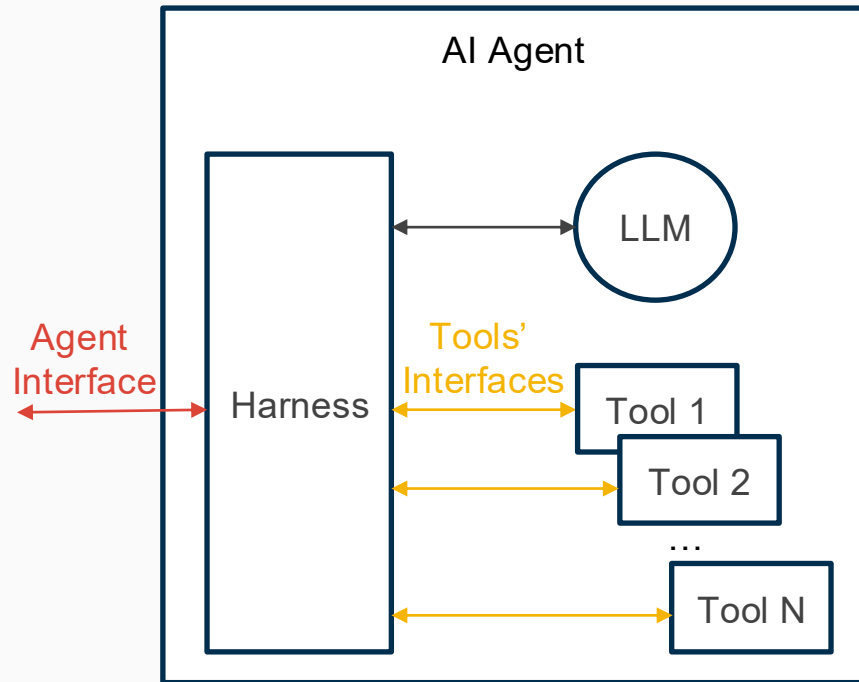
```
form: 1099-NEC  
reason: self_employment_income  
next_step: file_schedule_c
```

B

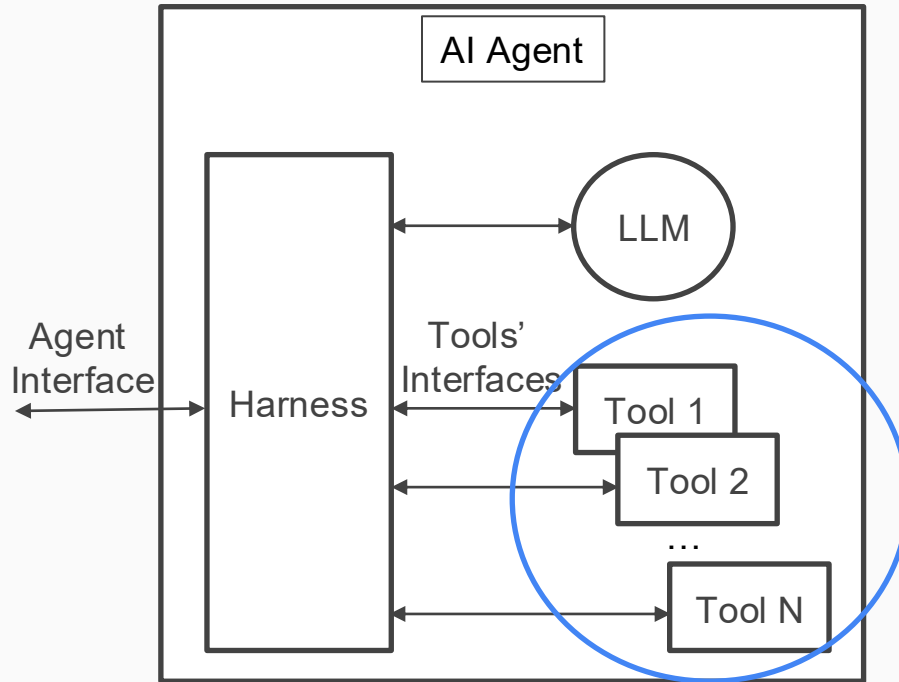
Here is the list of possible explanations of your tax situation, possible deductions, audit risk, filing strategy, and inferred financial profile ...

What is an AI agent?

- AI Agent = Harness + LLM
- Harness: deterministic runtime around the LLM
- driving the LLM, executing tools on it's behalf, etc.
deterministic runtime around the LLM
- Tools can be things like RAG, memory, REST API calls, JSON-RPC with fuzzy inputs and outputs
 - LLMs are exceptionally good in working out the fuzzy things
- Tools interfaces today are under-specified
 - Whenever possible, move away from the fuzzy Tools



Empower the Harness

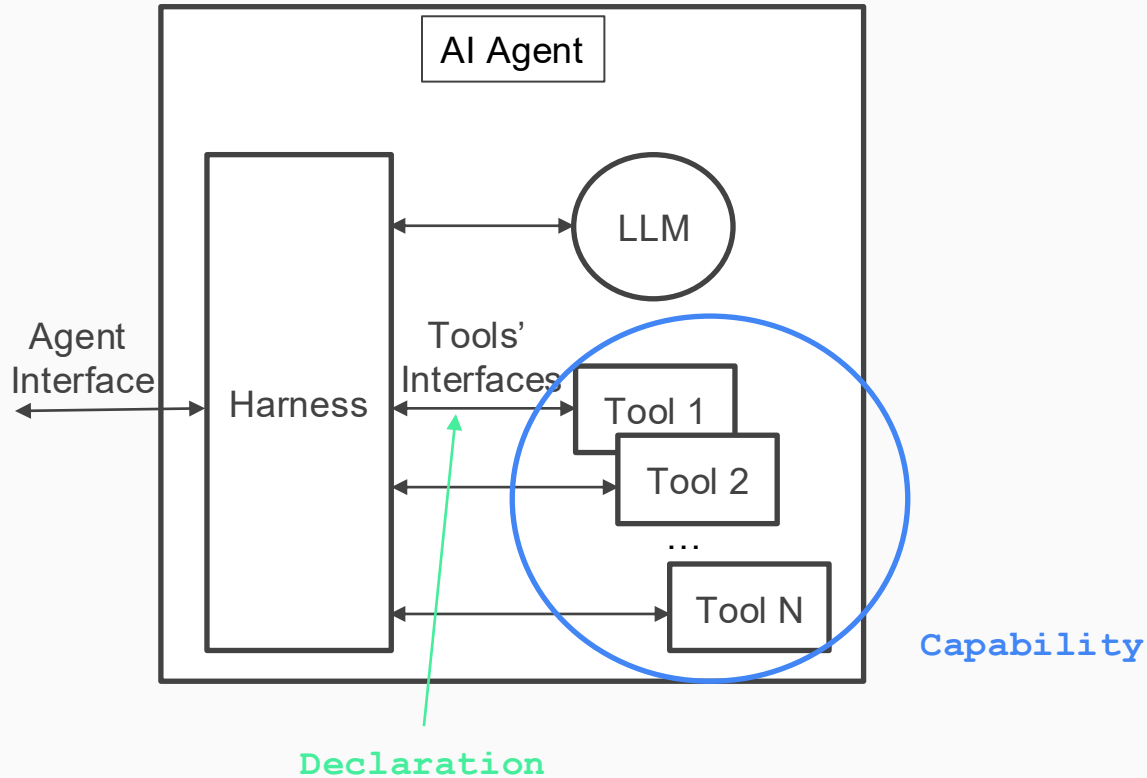


If the Capability is providing just about any output possible, with just any error paths possible, the Harness can do little before passing this to the LLM.

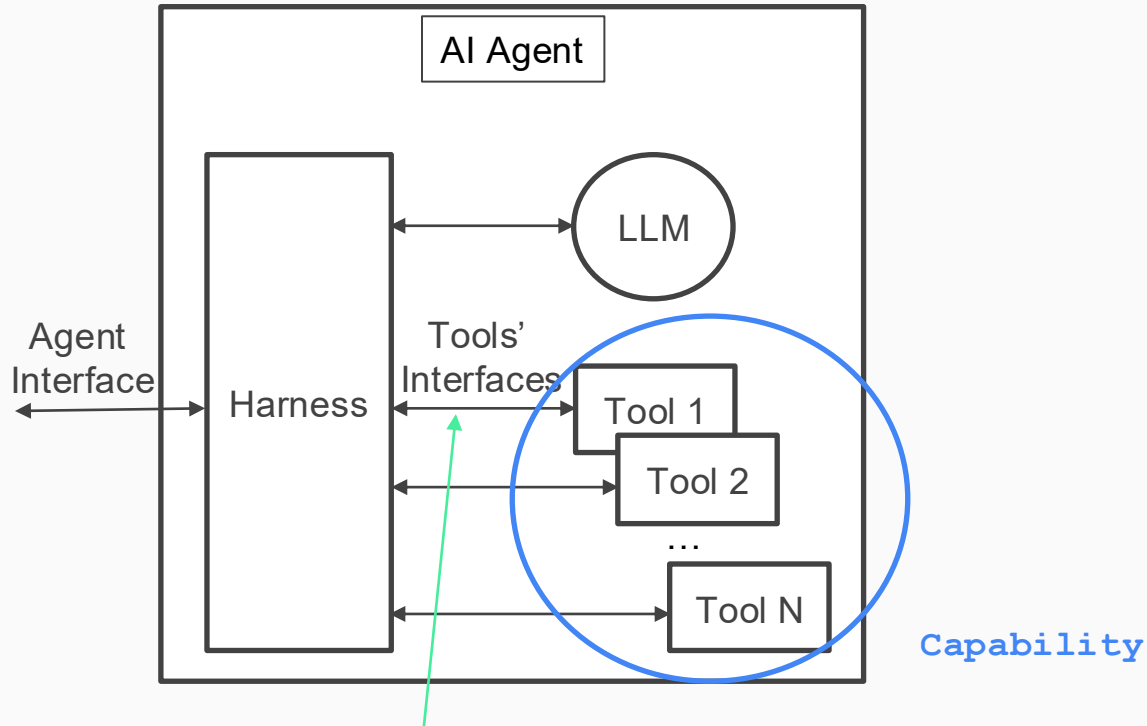
Using an LLM to evaluate if there is an error in the output, or how to handle an error – is non-deterministic.

Capability

Empower the Harness



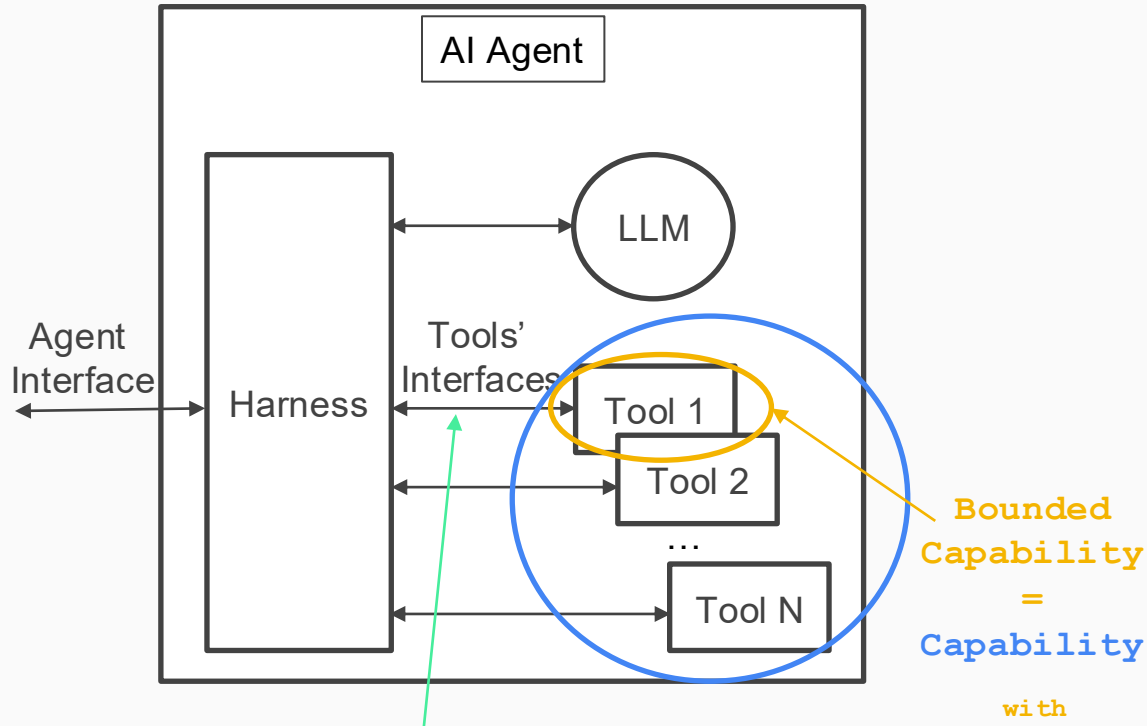
Empower the Harness



Declaration + Commitments = Contract

draft-pelov-bounded-agent-capabilities

Empower the Harness



Declaration + Commitments = Contract

draft-pelov-bounded-agent-capabilities

For inputs,
outputs and
errors

Bounded Capabilities

- Some Capabilities cannot be Bounded
 - But some CAN -> and that opens a lot of possibilities
 - Boundedness avoids discovering things, which can be defined compile-time
- Link between Bounded Capabilities and Agent Authorization
- Gaps today in MCP (which are interesting?)
 - boundedness is unsigned: caller cannot know if checks are possible
 - Conformance Is Self-Certified: no external confidence
 - Declarations Are Not Pinned : a contract can drift
 - Errors Are Untyped by Design : harness needs to react to non-deterministic errors

What's next?

- There is a link between the outer (e.g. AI Agent) interface and the inner (e.g. interface to the tools)
- Some of the gaps may be a non-issue (e.g. certifying by a third party)
- An example solution is described in draft-pelov-rich-architecture
 - RESTful Interface for the Control-plane of Harnesses